

Raunak Farhaz



DOB - 09 January 1997 | **Nationality** - Indian | **Current status**: Independent researcher

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Education

PhD (Dr. rer. nat.), Humboldt-Universität zu Berlin

2021 – 2026

Fully funded by DAAD doctoral research scholarship

Magna cum laude (defended on 21 April 2026)

Thesis: *Multiresolution Analysis in Extreme Magnetic Fields*.

Supervisor: PD Dr. Florian Bischoff

MSc Chemistry, Jadavpur University

2018 – 2020

Specialisation in Physical Chemistry

Thesis: *Theoretical Study of Adsorption of Carbon Monoxide on ZnO Clusters*.

Supervisor: Prof. Dr. Kalyan Kumar Das

BSc Chemistry, Jadavpur University

2015 – 2018

Research Experience

Doctoral researcher

Oct 2021 – Apr 2026

Humboldt-Universität zu Berlin, Germany | PD Dr. Florian A. Bischoff

- Investigated multiresolution analysis (MRA) for quantum chemistry in strong magnetic fields
- Quantified Gaussian basis-set errors against MRA results at complete basis set limit.
- Implemented a Monte Carlo algorithm with MRA Hartree-Fock to optimise He_n cluster geometries.
- Published the basis-set comparison results as first author in *Journal of Chemical Physics* (2025).

Visiting researcher

Sep 2023 – Oct 2023

UiT The Arctic University of Norway, Tromsø | Prof. Dr. Luca Frediani

- Interoperability of MADNESS and MRCPP (multiwavelet codes for quantum chemistry).

Research intern

Aug 2020 – Feb 2021

Humboldt-Universität zu Berlin | PD Dr. Florian A. Bischoff | Remote

- Studied quantum model systems using multiresolution analysis and Wolfram Mathematica.

Research intern

Apr 2018 – Sep 2018

Indian Association for the Cultivation of Sciences (IACS) | Prof. Dr. Abhijit Kumar Das

- Conducted a computational study of Al-S bond breaking with hydrogen evolution.

Selected Funding and Awards

1. Deutscher Akademischer Austauschdienst (DAAD) Doctoral Programmes

2021 – 2025

Research Grants – Doctoral Programmes in Germany, 2021/22. (PID - 91803487)

Fully funded the doctoral research stay at Humboldt-Universität zu Berlin, Germany.

2. DST INSPIRE Summer Research Internship Program

Apr 2018 – Sep 2018

Financial support for pursuing the research internship at IACS, India.

3. DST INSPIRE Scholarship for Higher Education

2015 – 2020

Scholarship for supporting BSc and MSc studies in natural (mathematics and physical) sciences.

Teaching and Mentoring

Guest lecturer, Mathematical Foundations

Oct 2025 – Nov 2025

International University of Applied Sciences, Berlin, Germany.

List of Publications

- Sardar, R.; Banik, R.; **Farhaz, R.**; Ghosh, S. "Drug-Induced Micelle-to-Supramicelle Transition in Cetyltrimethylammonium Bromide (CTAB): Comparative Study with Acetylenic Gemini Surfactant via Experimental and DFT Approaches" *Langmuir*, 42(28), 20765–20781. [2026]
[doi:10.1021/acs.langmuir.6c02934](https://doi.org/10.1021/acs.langmuir.6c02934)
- Farhaz, R.**; Bischoff, F. A.; Blaschke, S.; Stopkowicz, S. "Quantification of the basis set error for molecules in strong magnetic fields and general orientation" *Journal of Chemical Physics*, 163(3), 034308. [2025]
[doi:10.1063/5.0274736](https://doi.org/10.1063/5.0274736)

Technical Skills

Programming	Python, C++, Julia, Fortran, Wolfram Mathematica
Data analysis	NumPy, SciPy, pandas, Matplotlib
Quantum chemistry softwares	Gaussian, ORCA, PySCF, MADNESS, MRCPP/MRChem, ElemCo.jl, HUMMR
HPC stack	Linux, SLURM/PBS, MPI/OpenMP, Git, Make/makefile, CMake, GitHub Actions/GitLab CI, Snakemake, Fortran Package Manager (FPM), conda/uv/pip
Additional	Microsoft 365/Libreoffice, LaTeX, Typst, Quarto, HTML/CSS

Scientific Meetings

Year	Location	Title	Type	Poster/Talk ¹	Role ²
2025	Oslo, Norway	World Association of Theoretical and Computational Chemists (WATOC)	Conference	Poster	Attendance
2025	Berlin, Germany	Quantum Bio-Inorganic Chemistry (QBIC) Conference	Conference	NA	Organization
2025	Berlin, Germany	LOBSTER School in Chemical Bonding Analysis	School/Workshop	NA	Attendance
2025	Berlin, Germany	Numerical Methods in Quantum Chemistry (NMQC)	Conference	Poster	Organization
2023	Tromsø, Norway	Hylleraas annual meeting	Meeting	Talk	Attendance
2023	Tromsø, Norway	Numerical Methods in Quantum Chemistry (NMQC)	Conference	Poster	Attendance
2022	Heidelberg, Germany	Symposium for Theoretical Chemistry conference (STC)	Conference	NA	Attendance
2022	Paris, France	DFTK Workshop	School/Workshop	Poster	Attendance
2020	Kolkata, India	Chemical Sciences: Today and Tomorrow, Center for Advance Studies 2 program, University Grants Commission	Conference	NA	Attendance
2019	Kolkata, India	Chemical Sciences: Today and Tomorrow, Center for Advance Studies 2 program, University Grants Commission	Conference	NA	Attendance

¹NA = Not Applicable²Attendance = only participating

Social Activities

Founder and organiser, ChemistryCoffeeTalks

2022 – 2023

Monthly flash-talk series providing PhD students with a forum to present their research and network with peers, funded by **Humboldt Innovation Forum**, at the Humboldt-Universität zu Berlin, Germany

Languages

Languages	Bengali	English	German	Hindi
Level	Native	C2	B2	C2

References

Name	Affiliation	Role	Email
PD Dr. Florian A. Bischoff	Humboldt-Universität zu Berlin	PhD Supervisor	florian.bischoff@hu-berlin.de
Prof. Dr. Luca Frediani	University of Tromsø	Collaborator	luca.frediani@uit.no
Dr. Erik Tellgren	University of Oslo	Collaborator	e.i.tellgren@kjemi.uio.no
Prof. Dr. Trygve Helgaker	University of Oslo	Collaborator	trygve.helgaker@kjemi.uio.no
PD Dr. Denis Usvyat	Humboldt-Universität zu Berlin	Mentor	denis.usvyat@hu-berlin.de